MATH 3603–Homework 5

Due: 8-May-2026

1. Determine the period of all states in all Markov chains below

$$P_1 = \begin{bmatrix} 0 & \frac{1}{3} & \frac{2}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{4} & \frac{3}{4} & 0 \end{bmatrix}, \qquad P_2 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ \frac{1}{3} & \frac{2}{3} & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$P_3 = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 & 0 \\ \frac{1}{3} & 0 & \frac{2}{3} & 0 & 0 \\ 0 & 0 & 0 & \frac{2}{3} & \frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

2. A DNA nucleotide has any of 4 values. A standard model for a mutational change of the nucleotide at a specific location is a Markov chain model that supposes that in going from period to period the nucleotide does not change with probability $1 - 3\alpha$, and if it does change then it is equally likely change to any of the other 3 values, for some $0 < \alpha < \frac{1}{3}$.

(a) Compute the n-step transition probability from 1 to 1, i.e, p_{11}^n.

(b) Determine the long run proportion of time the chain is in each state. (*Hint: "long run" means limiting.)*

3. Consider three urns, one colored red, one white, and one blue. The red urn contains 1 red and 4 blue balls; the white urn contains 3 white balls, 2 red balls, and 2 blue balls; the blue urn contains 4 white balls, 3 red balls, and 2 blue balls. At the initial stage, a ball is randomly selected from the red urn and then returned to that urn. At every subsequent stage, a ball is

randomly selected from the urn whose color is the same as that of the ball previously selected and is then returned to that urn. In the long run, what proportion of the selected balls are red? What proportion are white? What proportion are blue?

4. Each morning an individual leaves his house and goes for a run. He is equally likely to leave either from his front or back door. Upon leaving the house, he chooses a pair of running shoes (or goes running barefoot if there are no shoes at the door from which he departed). On his return he is equally likely to enter, and leave his running shoes, either by the front or back door. If he owns a total of k pairs of running shoes, what proportion of the time does he run barefooted, in the long run?

5. Suppose that on each play of the game a gambler either wins 1 with probability p or loses 1 with probability $1-p$. The gambler continues betting until she or he is either winning n or losing m.What is the probability that the gambler quits a winner?

6. For the Markov chain with states 1, 2, 3, 4 whose transition probability matrix P is as specified below find f_{i3} and s_{i3} for $i = 1, 2, 3$.

$$P = \begin{bmatrix} 0.4 & 0.2 & 0.1 & 0.3 \\ 0.1 & 0.5 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.2 & 0.1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

7.: For a branching process, calculate the probability of ultimate extinction π_0 when (i) $P_0 = 1/4, P_2 = 3/4$; (ii) $P_0 = 1/4, P_1 = 1/2, P_2 = 1/4$; (iii) $P_0 = 1/6, P_1 = 1/2, P_3 = 1/3$, where P_k's describe the probability mass function of $\xi_{n,i}$, the number of the offspring of an individual.